

Soil compaction: a review of past and present techniques for investigating effects on root growth

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Abstract

Soil compaction has been known to affect root growth for millennia. Root growth in natural soils is complex and soil compaction induces several stresses which may interact simultaneously, including increased soil strength, decreased aeration and reduced hydraulic conductivity. Yet, moderate soil compaction offers some benefits to growing roots by increasing root–soil contact so they can extract adequate resources. Until now, improving our understanding of the specific responses of roots to below-ground stimuli has been difficult. However, the advent of new technologies and practices, including X-ray computed tomography, to provide non-destructive, three-dimensional images of root systems throughout the plant's lifecycle now allows the responses of roots encountering changes in their physical, chemical or biotic environment to be established directly and non-invasively. Previous destructive methods, such as root washing, were incapable of identifying and characterising fine root architectural characteristics as these are inextricably linked to the composition of the soil matrix. X-ray computed tomography coupled with genetic approaches will provide a more comprehensive appreciation of the effect of soil compaction on root growth, and the knowledge required to generate improvements in plant breeding programmes and crop husbandry.

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INTRODUCTION

The impact of soil compaction on plant growth has been acknowledged for centuries. Unger and Kaspar¹ referred to the principles of good soil management for optimum crop growth advocated by Cato the Elder (234–149 BC) and, more recently, McGarry² described soil compaction as the most serious environmental problem caused by conventional agriculture. It is widely accepted that the main causes of the recent widespread increase in soil compaction are the intensification of cropping systems and the evolution of agricultural machinery.^{3,4} Thus, as agricultural methods have become ever-more intensive over increasingly large areas of farmland, the machines used have become heavier, with the result that almost every model of tractor generates pressures exceeding the recommended maximum to avoid soil compaction.⁵ Surface compaction results mainly from the contact pressure of agricultural equipment with the ground, whereas axle load is primarily responsible for subsoil compaction.⁶ Soils with high bulk densities may also form naturally, such as the duplex soils found in Western Australia.⁷ Properties such as texture, clay mineral type and organic matter content all affect the ability of soils to withstand compaction. Moisture content also has a significant influence as saturated soils are less able to withstand the loading force of agricultural equipment in particular.⁴ However, the increased power and performance of modern agricultural vehicles allows them to operate under wet conditions, increasing the likelihood of compaction.

Compaction influences soil strength, aeration and water flow, creating inter-related stresses which may act simultaneously to influence root growth and distribution. As compaction alters the physical, chemical and biological characteristics of soil,

studies of the specific consequences for root growth are difficult, especially as the severity of compaction may be described using several parameters, including the number of tractor passes, penetrometer resistance, dry bulk density and total porosity.⁸ However, measurements of dry bulk density and porosity may be of limited value when comparing soils of differing type and texture. This led Hakansson and Lipiec⁹ to propose that the degree of compactness, which is virtually independent of soil composition, should be used to characterise compaction and provide directly comparable values. Other proposed approaches include the Proctor compaction test,¹⁰ the least limiting water range (LLWR)¹¹ and the S-theory.¹² Bengough *et al.*¹³ concluded that, in terms of measuring soil physical quality, the S-theory may be appropriate because it requires fewer parameters than LLWR, although the latter may define the physical stresses experienced by plants growing in compacted soil more effectively. However, the S-theory requires further testing and validation to substantiate this claim. The various parameters used to describe soil compaction highlight its complex nature and the need for further focussed research.

Extreme soil compaction may have major impacts on root growth and crop yield, although its initial effects such as stomatal closure and reduced shoot growth are not immediately obvious.¹⁴ For this reason, the effects of compaction on growth are often

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Table 1. Maximum axial and radial root growth pressures (from Whalley and Dexter²⁶)

Authors and year of publication	Plant species	Axial/radial pressure	No. of observations	Mean max. pressure (kPa)	Range of values (kPa)
Pfeffer (1893) (cited by Gill and Bolt (1995))	<i>Faba vulgaris</i>	Axial	6	1082	704–1936
	<i>Faba vulgaris</i>	Radial	6	509	390–611
	<i>Zea mays</i>	Axial	4	1451	953–2494
	<i>Zea mays</i>	Radial	1	659	–
	<i>Vicia sativa</i>	Axial	2	1080	826–1333
Taylor and Ratliff (1969)	<i>Pisum sativum</i>	Axial	63	1300	600–2600
	<i>Gossypium hirsutum</i>	Axial	64	940	600–1600
	<i>Arachis hypogaea</i>	Axial	71	1150	500–2000
Eavis <i>et al.</i> (1969)	<i>Pisum sativum</i>	Axial	290	1150	700–1600
	<i>Gossypium hirsutum</i>	Axial	242	888	400–1300
Aggarwal and Prihar (1975)	<i>Cicer areatinum</i>	Axial	10	450	120 (SD) ^a
	<i>Zea mays</i>	Axial	10	1090	187 (SD) ^a
Misra <i>et al.</i> (1986)	<i>Gossypium hirsutum</i>	Axial	37	497	–
	<i>Pisum sativum</i>	Axial	13	289	–
	<i>Helianthus annuus</i>	Axial	29	238	–
	<i>Pisum sativum</i>	Radial	70	900	–

^a Standard deviation of values.

difficult to quantify. This paper reviews the effects of compaction on root growth and explores new technology which will greatly aid further research into the consequences of compaction for root growth. In a recent review of soil compaction and land management, Batey¹⁵ stated that using methods 'blindly without knowledge of what lies beneath' will result in inadequate and uninterpretable information. New techniques which provide greater insight into how soil compaction affects root architecture and the physiological responses of plants must be developed and exploited to enhance crop breeding initiatives, so that knowledge gained in the laboratory can be applied under field conditions.

ROOT RESPONSES TO SOIL COMPACTION

Soil strength and root growth

Soil strength is determined by its density and moisture content and increases as soils dry,¹⁶ increasing resistance to root growth. Soil strength is extremely important as Whalley *et al.*¹⁷ reported a link between soil strength and effective stress with regard to the ability of roots to penetrate unstructured soils at matric potentials below –250 kPa. It is difficult to separate the effect of compaction on root growth from that of soil moisture status, as the ease of water uptake is reduced in compacted soil, perhaps explaining the associated increase in the number of root hairs. Importantly, Whalley *et al.*¹⁸ concluded that soil strength may limit crop productivity irrespective of soil moisture status and suggested that this was the main abiotic stress factor determining yield; they suggested that soil strength may provide a more reliable prediction of yield in winter wheat than matric potential or soil moisture content. Roots may also change their architecture in response to below-ground environmental conditions as mechanical stress has been shown to elicit lateral root generation in *Arabidopsis thaliana*, whereby physical bending of roots results in a lateral root forming on the convex side of the curve, enabling roots to manoeuvre around obstacles.¹⁹ This mechanism would be extremely useful in avoiding localised areas of acute soil compaction.

Bengough *et al.*²⁰ stated that the maximum soil strength which roots can overcome is determined by the maximum turgor generated within the root elongation zone and the shape and frictional characteristics of the root tip. Studies of changes in turgor in impeded roots have provided conflicting results as some showed that turgor pressure increases in impeded roots,²¹ whereas increases in osmotic potential in the apical cells of impeded roots observed in other studies imply a decrease in root turgor.^{22,23} This apparent variation in response may reflect differences in measurement methodology or the level of impedance employed.¹⁶ It is, however, widely accepted that the main driver of cell enlargement in plants is wall relaxation, causing at least a temporary reduction in turgor pressure.²⁴

Root resistance to mechanical impedance

Effective root growth requires a continuous network of appropriately sized soil pores. Gregory⁷ suggested that the minimum pore size through which roots may penetrate is 10 µm, and that few species have roots of smaller diameter. If a continuous pore structure of appropriate size is not present, the root tip must exert sufficient force to displace soil particles and create a channel.⁷ Pfeffer²⁵ was the first to measure root pressure directly by encasing roots in plaster of Paris and determining the pressure exerted using a calibrated spring; he found that root extension ceased under conditions of high mechanical resistance.²⁶ After testing several approaches, Clark *et al.*²⁷ concluded that the shear beam apparatus was the most accurate for measuring root pressures *in situ* in pea (*Pisum sativum* L.); this apparatus is highly sensitive and capable of measuring forces as low as 0.005 N. Bengough *et al.*²⁰ concluded that a probe can be inserted into soil, ideally at a rate similar to typical root elongation rates, and then, by subtracting the frictional resistance, a good estimate of the absolute resistance experienced by roots can be obtained.

Although some reports suggest that root pressure varies between species (Table 1^{26,28–32}), Clark *et al.*³³ observed similar maximum axial growth pressures in the roots of monocotyledonous and

Table 2. Relation between soil bulk density, penetration resistance and mean root diameter (from Kirby and Bengough³⁶)

Soil type	Bulk density (g cm ⁻³)	Penetration resistance (MPa)	Root diameter (mm ^a)
Sandy loam	1.4	0.67	1.0
	1.5	1.03	1.4
	1.6	2.02	1.6
Clay loam	1.3	0.74	1.0
	1.4	1.05	1.1
	1.5	1.52	1.2

^a Approximate root diameter 6 mm behind the tip.

dicotyledonous species; they concluded that differences in root growth pressure were not responsible for inter-specific differences in root penetration in compacted soil. Although experimental procedures are now available to measure resistance to root extension through soil under laboratory conditions, these remain difficult. 'Penetrometer resistance' is therefore commonly used to estimate the force required for roots to extend through soil as it is the axial component that dominates the magnitude of the force required.¹⁶ The resistance experienced by growing roots has been quantified using cone penetrometers to provide measurements of soil strength.¹⁶ However, previous studies have shown that cone penetrometers may over-estimate the root pressures required for elongation by between 2- and 8-fold.³⁴ Rotating penetrometers have been suggested to provide a more realistic measure of the root pressures required for extension growth,^{16,17} but there is little information regarding the relationship between penetration pressure and soil moisture content.³⁵ The latter study highlighted the need to distinguish between soil textural classes when estimating soil strength to improve predictions of their compressive properties.

Radial expansion of roots

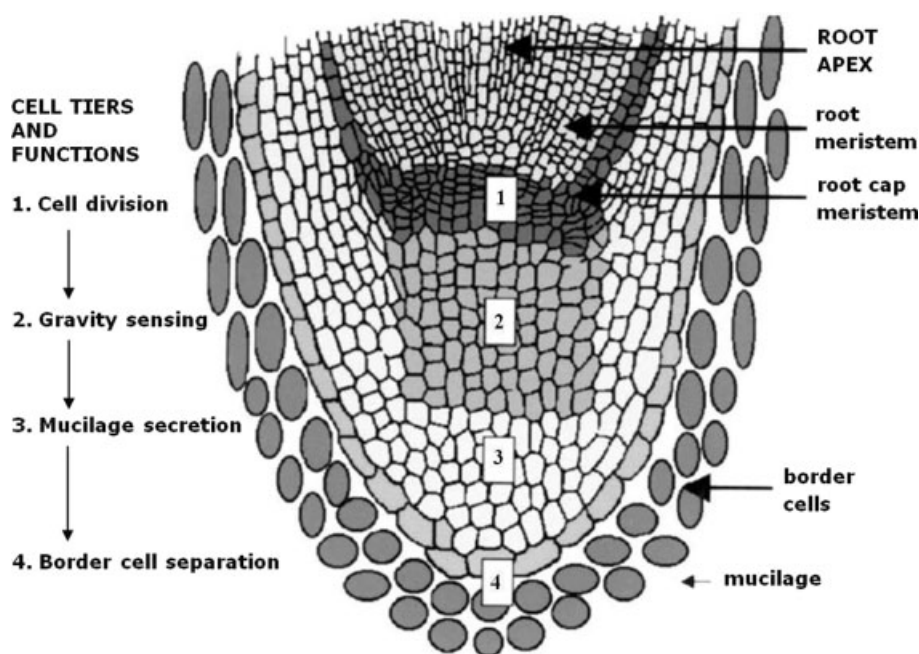
When roots encounter compacted soil, their initial response is an increase in radial expansion,¹³ resulting in shorter, thicker roots with an increased root hair density. Penetration resistance increases with increasing bulk density (Table 2), thereby increasing the diameter of pea roots.³⁶ Croser *et al.*²² reported that the cortical cells of roots grown in compacted soil were shorter and wider than in roots grown in loose soil, whilst Materechera *et al.*³⁷ concluded that the root diameter of seedlings grown in compacted soil increased by a factor of two due to the increased size of cortical cells and, occasionally, the presence of additional cell layers.³⁸

Root elongation rate

Passioura³⁹ reported that root growth slowed greatly when soil resistance reached 1 MPa and ceased entirely at 5 MPa. Bengough *et al.*¹³ concluded that a 2–5 day period was required for roots subjected to mechanical impedance to return to their unimpeded growth rate once they came in contact with uncompacted soil. Root elongation is slowed in compacted soil due to decreases in both cell flux (the rate at which new cells are added to a file) and final cell length.³⁸ The local strain rate is decreased towards the proximal end of the elongation zone, explaining why the elongation zone is shorter in impeded roots.³⁸ Differences between species in their ability to penetrate compacted soil may be attributed to variation in root diameter and the tendency of roots to deflect or buckle as opposed to differences in the pressure exerted by growing roots.⁴⁰ Interestingly, root elongation rates in high impedance sand were generally less affected in dicotyledonous than in monocotyledonous species,³⁷ perhaps explaining why dicotyledonous species are commonly used in crop rotations as they provide more effective penetration of compacted subsoil layers.⁴⁰

The root cap

Root cap cells have various functions (Fig. 1),⁴¹ although their main role is to protect the apical meristem from abrasion by

**Figure 1.** Root and apical meristem (from Hawes *et al.*⁴¹).

soil particles.⁴² Bengough *et al.*¹³ suggested that the root cap is important in determining the mechanical interaction between roots and the surrounding soil. They reported that, as the maximum stress encountered by roots occurs adjacent to the apex of the root cap, the root cap must influence the nature of soil deformation and frictional interactions between roots and soil. The root cap may assist the passage of roots by counteracting the resistance imposed by the surrounding soil.⁴³ Unlike other tissues, root cap cells are constantly renewed by meristematic divisions as the root cap is a persistent structural entity.¹⁶ Root cap cells are produced continuously and are sloughed off to provide a low-friction lining for the growing root at the soil-root interface to reduce frictional resistance.⁴² Iijima *et al.*⁴⁴ showed that removal of the root cap reduced root elongation by half in maize, an effect attributed to increased penetration resistance. Removal of the root cap also detaches mucilage-secreting cells (Fig. 1) which lubricate the passage of growing roots;³⁴ this would be expected to have a significant effect, as mucilage-secreting cells comprise almost half of the root cap volume in maize.¹⁶ For maize roots growing in a loamy sand soil, the decreases in penetration resistance resulting from the sloughing off of border cells and mucilage secretion were, respectively, 58% and 42%.⁴³

Root exudates

Root exudates secreted into the adjacent soil can be sub-divided into three categories: (1) mucilage, which is usually found at the root tips and consists of polysaccharides and uronic acids; (2) exudates from root hairs including amino acids, organic acids and simple sugars; and (3) cellular organic substances produced by senescence of the epidermis.⁴⁵ Gases are also released, including carbon dioxide and methane, although Gregory⁷ noted that some researchers^{46,47} do not define these as exudates as they are subsequently evolved to the atmosphere.

Czarnes *et al.*⁴⁸ reported that soil structure may be modified by repeated wetting and drying cycles and the release of exudates, which may alter the structural and hydraulic properties of the soil. Root exudates may also aid aggregation of soil particles, particularly in dry soil in which the viscosity of exudates is increased.⁴⁹ Czarnes *et al.*⁴⁸ suggested that exudates may stabilise soil structure in the rhizosphere by increasing the strength of the bonds between particles, whilst decreasing the wetting rate. Extracellular compounds from arbuscular mycorrhizal fungi (AMF) form hydrophobic coatings on soil particles which are also believed to increase soil stability.⁵⁰

Exudates are also important in maintaining root–soil contact in drying soils because the surface tension of the exudate decreases as the soil dries, increasing its ability to wet surrounding soil particles.⁵¹ Other studies suggest that the rhizosphere may be wetter than bulk soil despite relatively high water extraction rates⁵² due to the formation of a ‘coherent sheath of soil permeated by mucilage and root hairs’ known as the rhizosheath.⁷ Small quantities of water released from roots to the rhizosheath at night are re-absorbed during the following day.⁴⁹ The rhizosheath therefore has a significant influence on soil hydraulic properties and roots generally modify soil structure and hence its water retaining capacity. However, Whalley *et al.*⁵³ reported that root-derived mucilage did not affect the infiltration rate for water and proposed that the influence of mucilage on soil hydraulic properties was related to soil organic matter content.

PLANT RESPONSES TO SOIL COMPACTION

Resource capture

Besides aiding the physical penetration of roots, environmental sensing is another important role of the root cap as it is the first point of contact between growing roots and their immediate physical and chemical soil environment; their perception of and response to local environmental conditions is therefore vital.¹⁶ As the primary task of roots is to absorb and transport water and nutrients to the rest of the plant, the ability to sense where available supplies are located is vital. Hawes *et al.*⁴¹ described the root cap as responding to a ‘symphony’ of signals from the environment. Soil compaction affects resource capture as pore size distribution ultimately determines the quantity of water which soils can hold and for how long. At field capacity, clay soils have a much higher percentage water content expressed on a dry-weight basis than coarse sandy soils.⁵⁴ Soil structure therefore significantly affects the quantity of water available for uptake by roots and the possibility that roots may experience anoxic conditions resulting from waterlogging. The quantity of water held at field capacity is also important as there is very little movement of water in soil once field capacity has been reached.⁵⁴

Much information concerning the effects of compaction on nutrient uptake is contradictory; for example, Lindberg and Pettersson⁵⁵ showed that uptake rates were lower for impeded than for unimpeded roots, whereas Andrade *et al.*⁵⁶ found no restriction of nutrient uptake by impeded roots of sunflower. Masle and Passioura⁵⁷ found that nutrient uptake was independent of soil strength in 22-day-old wheat seedlings, at a time when they were still at least partially reliant on seed-derived nutrients; negative effects of soil strength on nutrient uptake might therefore only be recognised at later growth stages. Compaction alters the mean size of pores in the soil matrix and hence the connectivity of the pore network as macropores are preferentially compacted. Atkinson *et al.*⁵⁸ concluded that this has a significant effect on crop establishment by influencing the movement of water and solutes, biological activity and soil aeration. However, although the preferential compression of macropores during compaction might be expected to have little effect on soil microbial communities as these generally occupy the smaller micropores, Gregory *et al.*⁵⁹ reported smaller fungal yields in compacted soil, suggesting that detrimental effects of compaction on microbial communities may adversely affect root growth.

In most terrestrial plants, a proportion of the soil pore space must be air-filled to supply oxygen for root metabolism; if the soil is not adequately ventilated, the supply of oxygen and removal of carbon dioxide will be reduced, restricting root growth and impairing plant performance and productivity. Accumulation of carbon dioxide and lack of oxygen also reduces the permeability of the roots to water,⁵⁴ to the detriment of plant growth. Adequate soil aeration is also essential to maintain the soil microbial population. Aeration depends on pore size distribution, with key indices being air capacity and available water capacity.⁶⁰ Aeration is rarely a problem in sandy soils, whereas fine textured soils such as clays often experience anaerobic conditions as water drains more slowly from smaller pores.⁶¹

Root-to-shoot signalling in response to soil compaction

For a compound to be classified as a root-to-shoot signal, it must move acropetally *via* apoplastic or symplastic pathways and influence physiological processes in target organs remote from the site of synthesis.⁶² Russell and Goss⁶³ were amongst

the first to suggest that responses to mechanical impedance were hormonally induced, while Masle and Passioura⁵⁷ were the first to publish evidence that chemical signals were involved when they showed that responses to increased soil strength were consistent regardless of whether these resulted from soil drying or compaction. As soils dry, root-sourced signals are transported through the xylem to leaves, reducing water loss and leaf growth,⁶⁴ these workers also noted that, although the presence of chemical signals was not refuted, their identity remained controversial and that abscisic acid (ABA), pH, cytokinins, 1-aminocyclopropane-1-carboxylic acid (ACC), a precursor of ethylene, malate and other unidentified factors have all been implicated in root-to-shoot signalling.

ABA is widely recognised as a stress hormone involved in mediating responses to soil compaction and water stress.⁶⁵ The root tips usually contain the highest ABA concentrations due to the low vacuolisation of the cells and consequently high percentage of cytosol, the compartment where ABA is formed.⁶⁶ However, as the relative water content of roots must be reduced by 50% to stimulate ABA production, many reports have questioned the sensitivity of this response¹⁶. De Kroon and Visser³⁸ suggested that the associated increase in xylem sap pH in roots growing in drying soil might direct the low concentrations of ABA which are constantly present in plants to their sites of action. This suggestion originated from observations that increases in xylem sap pH can be detected within hours, whereas significant increases in root-sourced ABA may be detected only after several days.

ABA moves in the xylem from roots to the shoot to regulate transpirational water loss and leaf growth.⁶⁶ During periods of stress, ABA conserves water by closing stomata,⁶² as is clearly demonstrated by the inability of 'wilty' mutants of tomato with a reduced ability to synthesise ABA to close their stomata.⁶¹ Mulholland *et al.*⁶⁷ proposed that the increased concentration of magnesium in the xylem sap of plants grown on compacted soil may also induce stomatal closure. However, experiments by Holbrook *et al.*⁶⁸ involving reciprocal grafting of roots or shoots of ABA-deficient mutants of tomato onto wild type material suggested that stomatal closure does not depend on ABA production by the roots, indicating that another chemical signal may be responsible for the increased ABA synthesis in leaves which promotes stomatal closure; Neumann⁶⁹ suggested that small peptide signal molecules with hormone-like activity were responsible.

Hussain *et al.*⁶⁵ suggested that elevated ABA concentrations reduce leaf growth during periods of water stress, a view supported by other studies showing inverse correlations between leaf growth and xylem ABA concentrations in maize.⁶² By contrast, Mulholland *et al.*⁷⁰ found that leaf expansion in wild-type barley plants was unaffected at a bulk density of 1.6 g cm^{-3} compared to plants grown on uncompacted soil, but was greatly reduced in the ABA-deficient barley mutant, Az34. They concluded that ABA has a positive role in maintaining leaf expansion at this 'sub-critical' level of compaction. However, at a bulk density of 1.7 g cm^{-3} , leaf growth was inhibited in both genotypes, suggesting that the ability of ABA to ameliorate the impact of compaction is limited to the 'critical' level of 1.6 g cm^{-3} .^{65,70}

Hussain *et al.*⁷¹ concluded that ethylene is also involved in regulating the growth of tomato plants experiencing compacted soil using isogenic wild-type and mutant genotypes with a reduced capacity for ethylene synthesis. Plants were grown in soil columns vertically divided by a polythene membrane to prevent lateral movement of water, nutrients and roots; one side of the column contained moderately compacted soil

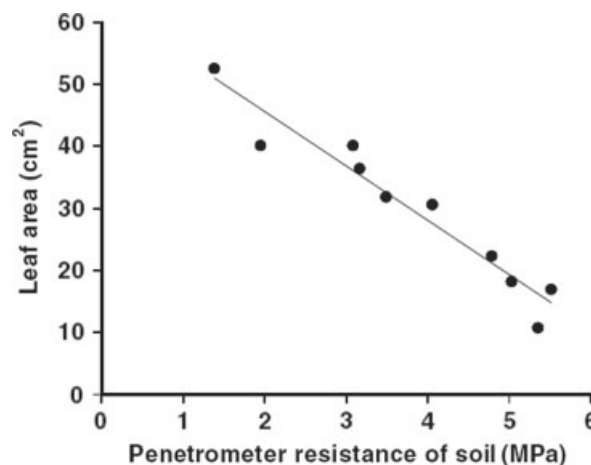


Figure 2. Effect of penetrometer resistance of soil on the leaf area of young wheat plants (from Masle and Passioura⁵⁷).

(1.5 g cm^{-3}) and the other uncompacted soil (1.1 g cm^{-3}). The results showed that the mutant genotype did not exhibit reduced growth in compacted soil, apparently because of its reduced capacity to produce ethylene. However, ethylene itself is unlikely to be the root-sourced signal transported in xylem sap³⁸ and its precursor, 1-aminocyclopropane-1-carboxylic acid (ACC), has received attention as a possible root to shoot signal.⁶² ACC was initially identified by Jackson⁷² as a key signal involved in mediating responses to waterlogging and is now recognised also to be a messenger for information regarding soil strength.⁷³ As the conversion of ACC to ethylene only occurs in oxygenated environments, ACC produced in the roots is converted to ethylene when it reaches the shoots.⁷⁴

Cross-talk between ABA and ethylene appears to be important in mediating plant responses to water stress, compaction and other abiotic stress factors such as elevated atmospheric ozone concentrations.^{16,75} Hussain *et al.*⁶⁵ reported an inverse correlation between ethylene evolution and xylem sap ABA concentration when tomato plants were grown in split pots containing soil of differing bulk density. Similarly, applications of fluridone, an aquatic herbicide which inhibits ABA biosynthesis, increased ethylene production and reduced root growth.¹⁶ This antagonistic relationship between ABA and ethylene may also mediate leaf growth when plants encounter compacted soil.¹⁴ Future studies of the role of cross-talk between ABA and ethylene in determining plant responses to abiotic stresses will be greatly facilitated by the existence of isogenic genotypes of important dicotyledonous and monocotyledonous crops such as tomato and barley which under- or over-produce ABA and ethylene.

Effects on yield

As agricultural machinery is a major cause of compaction, cultivation practices are important in determining soil structure and hence water, nutrient and oxygen supplies to roots through their influence on soil strength and pore dynamics, both of which influence seedbed performance and crop establishment.⁷⁶ However, contrasting responses have been reported, as Maduakor⁷⁷ found that compaction increased the surface area of cassava leaves but reduced the production of storage roots, whereas Masle and Passioura⁵⁷ reported leaf area decreased with increasing soil resistance (Fig. 2). Seed germination may also be affected, as Bassett *et al.*⁷⁸ found that compaction reduced the number and speed with

which roots of *Cordyline australis* (Forst.f.) seedlings penetrated the soil. Reductions in elongation rate resulting from difficulty in generating sufficient force for roots to displace soil particles and extend through the spaces created may pose serious problems in agriculture. Taylor and Ratliff⁷⁹ stated that the mean elongation rate necessary for satisfactory establishment of cotton and peanuts under typical Alabama conditions was 1.0 cm d^{-1} and suggested that crop establishment will be poor when soil strength is sufficient to reduce root elongation below this critical value, as access to water and nutrients will be restricted.⁴⁰

However, as noted previously, moderate compaction of the topsoil may be advantageous for some soil types, suggesting there is an optimum degree of compaction for crop growth.^{4,76,80} This may be beneficial by increasing water retention and providing the close contact between the roots of germinating seeds and seedlings and the adjacent soil necessary for good crop establishment.⁵⁸ Excessively loose soil may reduce crop yields, perhaps because roots grow preferentially through large pores, with the result that sufficient contact between roots and soil for effective extraction of water and nutrients is not attained.³⁹ Although the reduced growth of plants grown on loose soil may reduce the likelihood of lodging in cereals,⁸¹ several field studies have shown that excessively loose soil conditions may adversely affect crop growth, especially when the early part of the growing season is relatively dry.^{9,76} This effect may be attributable to the low unsaturated hydraulic conductivity of loose soils,⁸² coupled with the limitations imposed by poor root-to-soil contact on water and nutrient uptake.⁸³ The traditional use of agricultural rollers is therefore beneficial by creating mild top soil compaction which improves moisture retention and seedbed quality.

INNOVATIVE TECHNIQUES FOR STUDYING THE IMPACT OF COMPACTION ON ROOT GROWTH

X-ray computed tomography

Direct investigation of root systems growing in soil has previously been impossible as they are encased within the opaque soil matrix and cannot be visualised directly.⁸⁴ Destructive or invasive techniques, such as root washing, mini-rhizotrons and observations in two dimensions, including thin sections and electron microscopy,⁸⁵ have previously been the most commonly used methods. However, visualisation must be made in three dimensions to quantify parameters such as pore connectivity.⁸⁶ Recent studies have shown that X-ray computed tomography (CT), which is widely used in the medical sciences, can be applied for *in situ* studies of root systems in soil,⁸⁷ making this approach eminently suitable for studies of the impact of soil compaction on root architecture. X-ray CT uses X-rays to acquire cross-sectional image slices of an object which contain information regarding the attenuation of the X-rays; this is a function of the density of the sample material.⁸⁸ These image slices can be reconstructed to provide a three-dimensional (3-D) visualisation of the sample volume.

The use of X-ray CT to examine undisturbed roots systems growing in soil is relatively new innovation, although Watanabe *et al.*⁸⁹ first employed this approach to visualise the coarse storage roots of Chinese Yam (*Dioscorea oppositifolia* L.) in 1992. Jenneson *et al.*⁹⁰ used 3-D time-lapse images of growing roots of wheat (*Triticum aestivum* L.) seedlings to determine root length and volume, while Heeraman *et al.*⁸⁴ obtained 3-D images of intact root systems of bush bean (*Phaseolus vulgaris* L.) using X-ray CT, although only to a depth of 0.8 cm. Gregory *et al.*⁹¹ observed root

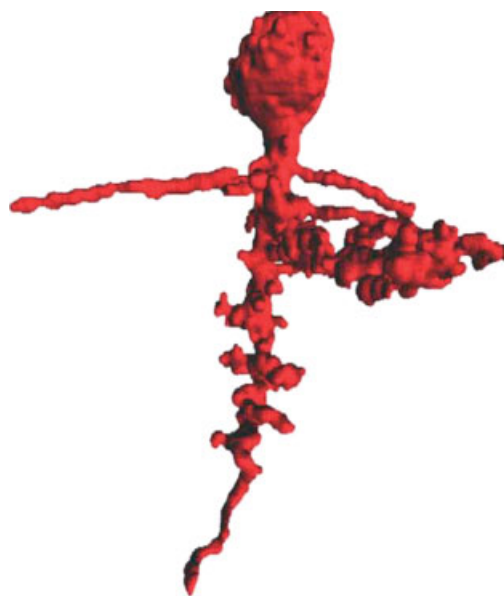


Figure 3. Three-dimensional reconstruction of chick pea (*Cicer arietinum* L.) root system 21 days after germination (from Perret *et al.*⁹²).

growth for pre-germinated wheat and rape seedlings transplanted into a sandy loam soil in 25 mm diameter plastic containers with an image resolution of $100 \mu\text{m}$. Similarly, Perret *et al.*⁹² used X-ray CT to visualise and quantify the roots of chick pea (*Cicer arietinum* L.), while Kaestner *et al.*⁹³ reconstructed the fine roots of two alder (*Alnus incana* L.) plants using X-ray micro-tomography. This approach was also used to measure root traits in barley (*Hordeum vulgare* L. ssp. *vulgare* and *spontaneum*).⁹⁴ McNeill *et al.*⁹⁵ visualised the effect of interfaces between high and low bulk density horizons on root architecture and highlighted how micro-tomography enables non-invasive and repeated analysis of interactions between roots and the soil microstructure.

Lontoc-Roy *et al.*⁹⁶ concluded that the combined use of high resolution CT scanning, computer programming and mineral soils with little organic matter (e.g. homogeneous sand and loamy sand) allowed accurate visualisation of undisturbed root systems. However, subsequent improvements in scanning technology mean that root distribution can now be visualised with a high degree of precision for most soil types; for example, the roots of *A. thaliana* L. can be visualised in heavy clay as modern scanners can achieve a spatial resolution of $<500 \text{ nm}$.⁸⁷

One of the main limitations of CT scanning for this purpose is the difficulty of extracting root systems from greyscale X-ray images due to the similarity in density of roots and adjacent moist soil. Until recently, this has usually been done by manual image processing, which is highly subjective. An effective automated process is therefore essential to extract root systems accurately from the soil matrix in CT images.⁹³ Several attempts to create root tracing software have been made based on various mathematical and computational algorithms.^{91,93,96,97} An example of an image created by 3-D root tracing produced by Perret *et al.*⁹² is shown in Fig. 3, although these workers admitted that some lateral roots were missing as the limited resolution of their CT scanner meant that fine root segments could not always be detected. Figure 4 shows the root system of a wheat (*Triticum aestivum* L.) seedling growing through a soil column which, was traced and extracted in three dimensions using the root tracing software RootViz[®] (in Tracy *et al.*⁸⁷). As the number of researchers using X-ray CT

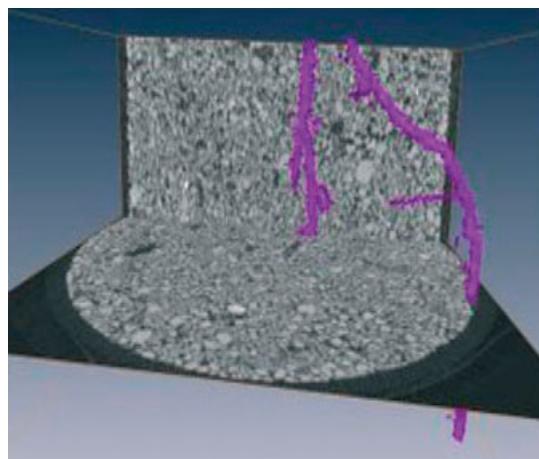


Figure 4. Cross section of a soil column (3 cm diameter) showing a root of *Triticum aestivum* L. traced in 3-D using Rootviz; roots are shown in purple (from Tracy *et al.*⁸⁷).

increases, the root tracing algorithms available will continue to improve in accuracy, as will their ability to characterise differences in the rooting pattern of fine lateral roots and atypical rooting directions.

Other three-dimensional visualisation techniques

Other approaches suited to the study of the impact of soil-borne stresses on root growth include gamma-ray computed tomography, neutron radiography, nuclear magnetic resonance (NMR) and magnetic resonance imaging (MRI). NMR uses magnetic field gradients to produce images showing the distribution of protons within the target object.⁹⁸ One of the main applications of NMR in soil science has been to gain information regarding pore size distribution, although this approach has not yet been widely adopted in soils research, perhaps due to a lack of theoretical knowledge by users and suitable training opportunities.⁹⁸

Gamma-ray computed tomography and neutron radiography operate in a similar manner to X-ray CT except that the radiation source differs; for example, the radioactive source of gamma rays in the system described by Pires *et al.*⁹⁹ is americium-241. Menon *et al.*¹⁰⁰ explained that neutron radiography shows different characteristics within the target object compared to X-ray CT as the former utilises neutrons, which interact with the nuclei of atoms, whereas X-rays interact with the electronic shells of atoms. Neutron radiography can visualise soil water content much more effectively than X-ray CT and has been elegantly used to investigate water dynamics within the rhizosphere.¹⁰¹

A major disadvantage of NMR and MRI compared to X-ray CT is that most soils contain iron or manganese ions, which are paramagnetic,⁸⁴ limiting studies to roots with diameters <1 mm.⁹² Jenneson *et al.*⁹⁰ concluded that X-ray CT is ideal for 3-D visualisation of soil because its ferrous-rich nature renders magnetic techniques impractical and, as soil is a highly attenuating material, the flux provided by gamma radiation is insufficiently strong.

The use of genetic techniques to establish root-sourced signals

As noted previously, understanding is still limited regarding the signals involved in mediating plant responses to compaction and their modes of action. The use of mutant and transgenic genotypes will be important in elucidating the role of specific plant growth regulators and other signals in mediating physiological and growth responses to compaction.¹⁰² Many hormonal mutants of *A. thaliana* L. have been generated through collection of populations from the wild and various mutagenesis strategies.¹⁰³ Ethylene and abscisic acid (ABA) hormone mutants will be useful for studies of the effect of mechanical and hydraulic stress on root growth. Attempts to produce entirely ABA-free mutants have failed as it would be impossible for such plants to survive in terrestrial environments where numerous stress factors may be experienced simultaneously,¹⁶ demonstrating the key role of ABA in regulating responses to abiotic stress. However, *A. thaliana* mutants deficient to varying degrees in their ability to produce ABA and ethylene are available^{104–108} (Table 3) and may have an important role in future studies to elucidate the role of these plant growth regulators in mediating responses to soil compaction.

Future studies of the role of root-to-shoot signalling in mediating the impact of soil-borne stresses must also include important crop species to gain a full understanding of the mechanisms involved and how these may be manipulated to enhance crop performance and improve food security. One crop species offering a wide range of ethylene and ABA mutants is tomato (*Lycopersicon esculentum* Mill.). The use of mutant tomato varieties such as 'Never ripe' (Nr), in which the perception of ethylene is blocked,¹⁰⁹ would provide further insight into the impact of compaction on root growth and the role of long distance signals in regulating plant development.

As plants are sessile, they must alter their gene expression dramatically to adapt to and tolerate environmental change.¹¹⁰ Traditional methods of determining gene expression were previously restricted to analysing single genes at any one time, whereas new techniques allow simultaneous analysis of the expression of all genes within the genome.⁵⁴ The application of such techniques should provide a more precise understanding

Table 3. *Arabidopsis thaliana* hormone mutants (adapted from Koncz *et al.*¹⁰⁴)

Locus	Name	Phenotype	Reference
<i>aba</i>	ABA deficient	Deficient in abscisic acid	Koornneef <i>et al.</i> (1982)
<i>abi1</i>	ABA insensitive	Increased resistance to ABA (dominant)	Koornneef <i>et al.</i> (1984)
<i>abi2</i>	ABA insensitive	Resistant to ABA	Koornneef <i>et al.</i> (1984)
<i>abi3</i>	ABA insensitive	Resistant to ABA	Koornneef <i>et al.</i> (1984)
<i>etr</i>	Ethylene insensitive	Ethylene insensitive (dominant)	Bleecker <i>et al.</i> (1988)
<i>abi5</i>	ABA insensitive	ABA insensitive	Wang <i>et al.</i> (2007)
<i>abi8</i>	ABA insensitive	ABA insensitive	Wang <i>et al.</i> (2007)
<i>ein2</i>	Ethylene insensitive	Ethylene insensitive	Wang <i>et al.</i> (2007)

of the influence of soil compaction on root and shoot growth and crop productivity. As compaction creates a range interacting abiotic stress factors, including increased soil strength and reduced oxygen supplies, the use of full gene expression techniques to provide transcript profiles may facilitate the identification and subsequent manipulation of specific genes involved in mediating responses to the external stimuli.

CONCLUSIONS

Soil compaction and its effects on root growth are complex, creating a range of interactive below-ground stresses which may not necessarily elicit obvious short term adverse effects on above-ground performance. Several studies have shown that there is an optimum level of soil compaction for root growth, but as natural soils are often highly heterogeneous, the roots of plants growing in such environments are likely to experience a range of soil bulk densities and may well respond to such local variation. However, previously unanswerable questions regarding effects on root architecture can now be addressed following the advent of new technologies such as X-ray CT scanning and genetic approaches. X-ray CT offers major advantages as it is non-invasive and can visualise soil structure in three dimensions. This opens up the ability to examine root growth over short time scales (i.e. hourly) to establish responses when roots encounter changes in their physical, chemical or biotic environment. With the incorporation of genetic approaches, a greater and more precise understanding of the consequences of soil compaction on root growth can be achieved.

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